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Research into AI Ethical & Responsible AI
Topic: How Should AIs Behave in Our Society?

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Executive Summary

This report examines the question of how artificial intelligence systems should behave in human society, with a focus on ethical responsibility and the challenges of AI bias from a global perspective. As AI systems increasingly influence decisions that affect individuals and communities, concerns around fairness, accountability, and the protection of human rights have become central to debates on responsible AI development and deployment.

To explore this topic, the report examines ethical principles drawn from established global frameworks, including UNESCO, the OECD, and the recent regulatory proposals such as Australia's mandatory guardrails for high-risk AI. By analyzing real-world case studies involving biased or unethical AI systems, the report identifies common causes of harm, such as biased data, a lack of transparency, and insufficient human oversight. The findings suggest that, although strong ethical principles exist, they need to be supported by enforceable governance mechanisms, inclusive design practices, and ongoing oversight to ensure that AI systems behave in ways that align with societal values.

1. Introduction

Artificial intelligence (AI) is a transformative technology that allows machines to perform human-like problem-solving tasks and enables businesses to make better decisions on a large scale (Amazon Web Services, n.d.). As it continues to evolve, AI increasingly raises more concern than excitement within the community. From its beginning as a theoretical foundation in the 1950s, to its breakthrough with deep learning in the 2010s, and the widespread deployment of powerful generative AI and transformer models today, AI has existed for only around 70 years, a brief moment in human evolution. But because of this rapid development, questions are raised about how AI systems should behave in human society.

In the community, ethical principles have long-guided behavior and decision-making across fields such as science, law, and technology (UNESCO, n.d.). However, AI systems cannot be directly compared to humans. Although AI systems are effective in data processing, predictive analytics, and automation, humans contribute more on creativity, ethical judgement, and interpersonal understanding (FiveRivers Technologies, n.d.). Despite lacking emotions or personal intentions, AI systems now influence decisions that shape access to employment, healthcare, and justice. As a result, public trust in institutions is increasingly connected to whether AI systems operate in ways that are perceived as fair, transparent, and accountable.

These concerns become more significant in sensitive domains where ethical issues such as discrimination, transparency, and accountability are already challenging. In real world, AI systems can appear objective while still producing unfair outcomes. This is partly because AI bias differs from human bias. Human bias can often be questioned through discussion and reflection, while AI systems may produce results through complex and opaque processes that make biased outcomes appear neutral or data-driven (Holistic AI, n.d.). When such systems are deployed widely, the consequences can affect entire populations rather than individuals.

These consequences are actually apparent in practical contexts. For example, in healthcare, Black patients experience significantly higher mortality rates for melanoma, with an estimated

five-year survival rate of only 70%, compared to 94% for White patients (The Atlantic, 2018). While this disparity is not caused by AI alone, it demonstrates how data-driven decision systems can reinforce existing inequalities if ethical risks are not carefully addressed. These challenges illustrate the importance of examining how AI systems are designed, trained, and governed, especially when they become more embedded in everyday decision-making across societies.

Because AI systems are now shaping outcomes that directly affect human well-being, it becomes significantly important that they operate in ways that protect human dignity, support fairness, and remain accountable to human oversight. To understand how this can be achieved requires examining what ethical behavior means for AI, why ethical failures occur, and how societies are currently attempting to address these risks at a global level.

This report adopts a qualitative research approach based on literature review and critical analysis. The study begins with the two base documents, Burton et al. (2017) and the Australian Government's proposal on mandatory guardrails for high-risk AI systems, and then expands to include peer-reviewed journal articles, policy papers, and reputable news sources. The selection of these sources aimed to achieve a balance between theoretical viewpoints and practical case studies. The findings are analyzed and synthesized to evaluate whether existing ethical principles are sufficient to guide AI behavior in society.

2. Findings & Analysis

2.1. Ethical Expectations for AI Behavior in Society, and Why Is That Challenging?

Understanding ethical behavior in AI first requires recognizing that AI systems do not make independent moral decisions. Their behavior instead emerges from data, design choices, and the goals defined by human developers and institutions. In this case, AI systems often reflect the values and assumptions embedded during development rather than creating their own ethical framework (Aaih, n.d.). In other words, when systems are trained on data shaped by historical inequality, they may reproduce and even amplify those patterns. Therefore, ethical behavior in AI depends not only on the technology itself, but on how serious fairness, dignity, and justice are prioritized throughout the development and deployment process.

One of the main reasons ethical failures continue to occur is the presence of bias within AI systems. While the technical details of AI development are beyond the scope of this writing, research consistently identifies bias arising from data, algorithm design, and human judgement (AAIH Singapore, n.d.). When training data lacks representation or reflects historical discrimination, AI systems may produce unfair outcomes. Algorithmic design can also generate unequal error rates between groups, even when overall accuracy appears acceptable.

For instance, in 2016, an investigation by ProPublica examined the COMPAS risk assessment tool used in U.S. courts to predict recidivism. The analysis found that Black defendants who did not reoffend were more likely to be incorrectly labeled as "high risk," while white defendants were more frequently misclassified as "low risk." Although the system demonstrated similar overall accuracy across groups, the unequal distribution of errors raised serious concerns about racial bias and competing definitions of algorithmic fairness.

These kinds of problems show that bias in AI is not only a technical issue. As Selbst et al. (2019) argue, AI systems should be understood as socio-technical systems that operate within broader social and institutional contexts. When complex social issues are simplified into technical models, important real-world factors can be overlooked. This means that even if an algorithm appears mathematically fair, it may still produce harmful outcomes because it does not fully account for the social environment in which it is used.

In addition, human assumptions influence how problems are framed, what variables are prioritized, and how outputs are interpreted. As a result, AI systems are not neutral technologies, but socio-technical systems shaped by human and institutional choices.

All together, these forms of bias demonstrate that ethical failures in AI do not come from isolated technical errors. They emerge from the interaction between social context, development decisions, and deployment environments. AI systems do not only assist decision-making. In many cases, they replicate and scale underlying assumptions across contexts. If bias begins in design and data, its effects intensify as AI technologies are deployed across different countries and cultures, transforming localized ethical concerns into broader global challenges. Because of this, ethical expectations for AI must go beyond technical performance and instead prioritize fairness, accountability, transparency, and sustained human oversight.

2.2. AI Bias as a Global Ethical Challenge

AI systems are now used widely across many areas of modern society, especially in sensitive and high-impact domains such as criminal sentencing, loan approval, personnel recruitment, and healthcare decision-making. In these contexts, AI tools are often considered efficient, data-driven, and objective alternatives to human judgement. However, this perception can be misleading, because when AI systems are trained on historical, incomplete, or socially skewed data, they may reproduce existing patterns of inequality rather than eliminate them.

Because of this, ethical concerns surrounding AI cannot be understood purely as technical problems. Instead, they emerge from the interaction between technology, social structures, and institutional decision-making processes. AI does not exist alone. It is created within human systems and later deployed back into those same systems. Thus, when bias exists in society, AI can unintentionally amplify it.

This challenge becomes even more complex when AI systems are deployed globally. Biased AI systems do not remain confined to the environments in which they were originally developed. As AI technologies spread across borders, they carry assumptions, values, and limitations shaped by their original cultural, economic, and legal contexts. For example, AI models in healthcare that are trained primarily using data from high-income countries may perform less accurately when applied to underrepresented populations in other regions. Instead of reducing inequality, these systems may unintentionally reinforce existing global health disparities (arXiv, 2024).

At the same time, societies are different in how they define fairness, accountability, and acceptable risk. In some areas, like the European Union, precautionary regulations and the protection of basic human rights are very important. This is reflected in initiatives such as the EU AI Act, which attempts to address ethical risks before systems are deployed. Other jurisdictions

may prioritize technological innovation and economic competitiveness, sometimes accepting higher uncertainty or potential risk in exchange for faster development. Because of this, the same AI system may be considered acceptable in one country while raising serious ethical concerns in another (Beyond Technology, 2024).

These differences create a major challenge for global AI governance. They raise an important question: should AI systems be guided by universal ethical principles, or should ethical standards be adapted to local cultural values and regulatory priorities? There is no simple answer. Nevertheless, the existence of this tension highlights how difficult it is to balance diversity, innovation, and responsibility at the global level.

Understanding this global dimension is essential, because ethical concerns are not theoretical. They appear in real-world situations where AI systems directly affect people's lives. To better understand how these risks manifest in practice, it is necessary to examine real cases where biased or unethical AI systems have caused harm.

2.3. Case Studies of Biased / Unethical AI

How does biased or unethical AI appear in practice? It is important to examine real-world cases rather than relying only on theoretical discussion. These cases demonstrate how AI systems can influence people's lives in direct and sometimes harmful ways, particularly in sensitive domains such as healthcare, employment, and digital platforms.

Earlier well-known examples such as the COMPAS criminal risk prediction system and Amazon's hiring algorithm illustrate how biased data can lead to unfair outcomes. The COMPAS system demonstrated unequal error rates across racial groups, whereas Amazon's recruitment model learned to prefer male candidates because it was trained on historical hiring data dominated by male applicants. These examples show how AI can reproduce existing social inequalities when bias exists in training data or system design.

However, ethical risks in AI are not limited to automated decision-making systems. As AI becomes more integrated into public-facing services, new types of harm have been introduced. One example is the chatbot "Tessa," developed by the U.S. National Eating Disorders Association. The chatbot was designed to provide safe support for vulnerable users. However, it was later found to provide advice encouraging restrictive eating and weight loss behaviors. This directly conflicted with medical guidelines and created risks for users who were already vulnerable (The New York Times, 2023). Although the system was not intentionally designed to cause harm, this case demonstrates how insufficient oversight and over-reliance on automation can lead to dangerous outcomes.

More recent ethical concerns have emerged with generative AI systems capable of creating highly realistic images. In early 2026, controversy surrounded Elon Musk's AI system Grok, which is integrated into the X platform. Reports indicated that the system could be used to generate sexually explicit deepfake images of real individuals, often with minimal restrictions. This created significant concerns related to privacy, consent, and online harm (BBC News, 2026). Consequently, certain nations, such as Malaysia and Indonesia, restricted access to this kind of chatbot due to concerns about social and ethical risks.

Across these examples, several common patterns can be observed. First, harm is often introduced unintentionally, even when developers aim to build helpful systems. Second, AI outputs often appear objective or technically authoritative, which can make harmful outcomes more difficult for users to question. Third, once deployed at scale, AI systems can spread harmful effects quickly and across large populations.

These repeated patterns help explain why governments and international organizations have started developing ethical frameworks to guide AI development and deployment. These frameworks attempt to reduce risks by addressing bias, misuse, and accountability before AI systems are widely deployed.

2.4. Global Approaches to Governing Ethical AI

Responding to the growing number of biased and unethical AI cases, international organizations and governments have begun developing ethical frameworks to guide how AI systems should be designed, deployed, and monitored. Organizations such as UNESCO and the Organization for Economic Co-operation and Development (OECD) aim to establish shared ethical foundations that can be applied across countries as AI systems increasingly operate on a global scale.

The OECD has proposed a widely referenced set of principles for responsible AI development (OECD, n.d.). These principles emphasize that AI should support inclusive economic growth, sustainable development, and overall human well-being rather than focusing only on efficiency or profit. Furthermore, the significance of upholding human rights and democratic principles, encompassing fairness, equality, and privacy, is underscored. These objectives are intended to be realized through the implementation of safeguards, transparency, and meaningful human oversight throughout the AI lifecycle.

Transparency and explainability are also important, as they allow individuals to understand how AI systems function and to challenge automated decisions when necessary. The OECD framework also emphasizes that AI systems must be technically robust, secure, and safe. This includes ensuring that systems can resist misuse and can be disabled if they pose serious risks. Finally, accountability is a core requirement, meaning organizations must take responsibility for AI-driven outcomes and actively manage risks related to bias and rights violations.

Despite these general principles, global governance takes on significantly different approaches. Some countries rely mainly on voluntary ethical guidelines and industry selfregulation. Others have introduced legally binding regulations that place clear obligations on developers and deployers of AI systems. These differences reflect national priorities, levels of technological development, and varying attitudes toward government regulation.

While these frameworks provide important guidance and direction, are these ethical principles sufficient to ensure that AI systems behave ethically in real-world environments?

2.5. Can Current Principles Ensure Ethical AI Behavior?

Ethical AI principles provide an important foundation for defining responsible AI behavior. They help organizations understand expectations related to fairness, transparency, accountability,

and safety. In this case, ethical principles serve as a starting framework for the responsible development of artificial intelligence.

However, ethical principles alone are not enough to guarantee ethical behavior in practice. A major limitation is that these principles are often broad and difficult to measure. Concepts such as fairness and transparency can be interpreted differently depending on context, making it challenging to translate high-level ethical values into specific technical requirements or regulatory standards.

As Barocas, Hardt, and Narayanan (2019) explain, fairness in machine learning does not have a single universally accepted definition. Different mathematical definitions of fairness may conflict with one another, which means that improving performance under one fairness metric can worsen outcomes under another. This tells us that fairness is not purely a technical objective, but a normative choice shaped by social priorities and policy decisions. Hence, ethical principles that call for "fairness" do not automatically provide clear guidance on how fairness should be measured or enforced in practice.

Another challenge is that ethical standards must evolve over time. As societies change and new AI capabilities emerge, new ethical risks also appear. This creates uncertainty about how ethical principles should be interpreted and updated to reflect new social realities and emerging human rights concerns.

Enforcement is another critical issue. In many cases, ethical AI guidelines rely on voluntary compliance. This creates a gap between policy intentions and real-world implementation. An organization may appear to follow ethical guidelines in documentation while still deploying systems that produce harmful outcomes due to data limitations, design decisions, or insufficient monitoring.

There is also a risk of organizations adopting ethical language without meaningful action. This is sometimes described as ethics washing, where ethical principles are used for reputation rather than accountability.

Because of these limitations, ensuring ethical AI behavior requires more than high-level principles. It requires strong governance structures, continuous monitoring, independent evaluation, and clear accountability mechanisms that can identify and address harm when it occurs.

3. Conclusion & Recommendations

The discussion across these sections shows that AI behavior is not determined only by technology. It is, in fact, shaped by the data it ingests, the design decisions made, the governance frameworks established, and the societal contexts within which these systems are utilized. Consequently, when bias exists in data or development processes, AI systems possess the capacity to replicate and exacerbate existing inequalities. Furthermore, the global deployment of such systems can spread across diverse societies, disproportionately impacting populations that are already vulnerable.

International ethical frameworks such as those developed by the OECD and UNESCO provide a shared foundation for responsible AI development. However, ethical principles alone cannot

ensure ethical behavior. Without measurable standards, enforcement mechanisms, and continuous oversight, there is a risk that ethical commitments remain theoretical rather than practical.

Therefore, AI should behave in ways that prioritize human dignity, fairness, transparency, and accountability. Achieving this requires cooperation between governments, industry, researchers, and civil society.

To move toward ethical AI behavior in practice, several actions are important. Independent audits should be required for high-risk AI systems before deployment. Organizations should provide clear documentation explaining how systems are trained and how they make decisions. Legal responsibility should be clearly defined so organizations are accountable when harm occurs. Human oversight should remain central, especially in decisions that significantly affect individuals. Finally, international cooperation is necessary to reduce global inequalities in AI governance and protection standards.

Lastly, ethical AI is not only a technical challenge. It is a social responsibility. The way AI behaves in society will depend on the collective choices made by developers, policymakers, and society as a whole.

4. Acknowledgements / Resources

In preparing this report, I consulted the two base documents provided in the assignment brief, including Burton et al. (2017) and the Australian Government's proposal on mandatory guardrails for high-risk AI systems. Additional academic journal articles, policy papers, and reputable news sources were used to support analysis and provide real-world case studies.

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